

TUPLE

```
In [7]: t = ()  
print(t)
```

```
()
```

```
In [8]: print(type(t))
```

```
<class 'tuple'>
```

```
In [9]: t1 = (10,20,30,40,40)  
t1
```

```
Out[9]: (10, 20, 30, 40, 40)
```

```
In [10]: len(t1)
```

```
Out[10]: 5
```

```
In [11]: t1.count(10)
```

```
Out[11]: 1
```

```
In [12]: t1.count(40)
```

```
Out[12]: 2
```

```
In [13]: t1.index(20)
```

```
Out[13]: 1
```

```
In [14]: l5 = ['a','b','c','d']  
l5
```

```
Out[14]: ['a', 'b', 'c', 'd']
```

```
In [15]: l5[1] = 10
```

```
In [16]: l5
```

```
Out[16]: ['a', 10, 'c', 'd']
```

```
In [17]: t2 = (100,3.4,'nit',True,1+2j,[1,2,3],(5,6,7))  
t2
```

```
Out[17]: (100, 3.4, 'nit', True, (1+2j), [1, 2, 3], (5, 6, 7))
```

```
In [18]: print(t)  
print(t1)  
print(t2)
```

```
()  
(10, 20, 30, 40, 40)  
(100, 3.4, 'nit', True, (1+2j), [1, 2, 3], (5, 6, 7))
```

```
In [19]: t1
```

```
Out[19]: (10, 20, 30, 40, 40)
```

```
In [20]: t1[0] = 1000 # Tuple is immutable  
t1
```

```
-----  
-----  
TypeError                                Traceback (most recent call  
last)  
Cell In[20], line 1  
----> 1 t1[0] = 1000 # Tuple is immutable  
      2 t1  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [23]: icici = (1234, 'asscscs', 23323232, 11211)  
icici
```

```
Out[23]: (1234, 'asscscs', 23323232, 11211)
```

```
In [24]: icici[0] = 123
```

```
-----  
-----  
TypeError                                Traceback (most recent call  
last)  
Cell In[24], line 1  
----> 1 icici[0] = 123  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [25]: t4 = t1 * 3
```

```
In [26]: t4
```

```
Out[26]: (10, 20, 30, 40, 40, 10, 20, 30, 40, 40, 10, 20, 30, 40, 40)
```

```
In [27]: t1[:7]  
t1
```

```
Out[27]: (10, 20, 30, 40, 40)
```

```
In [28]: t1[2:]
```

```
Out[28]: (30, 40, 40)
```

```
In [29]: t1[0:10:2]
```

```
Out[29]: (10, 30, 40)
```

Tuple creation

```
In [30]: tup1 = ()
```

```
In [31]: tup2 = (10,20,30,40)
```

```
In [32]: tup3 = (1.2,2.2,3.3,4.4)
```

```
In [33]: tup4 = ('a','b','c','d')
```

```
In [34]: tup5 = (1,2,3,(2,3,4),(4,5,6))
```

```
In [35]: tup6 = (10,'abhi',17,2)
```

```
In [36]: tup7 = ('ABHI',1,2,3,(10,30),(101,101),[101,1012],{1,2,6},(87,1,1))
```

```
In [37]: len(tup7)
```

```
Out[37]: 9
```

Tuple indexing

```
In [38]: tup7[0]
```

```
Out[38]: 'ABHI'
```

```
In [39]: tup7[1]
```

```
Out[39]: 1
```

```
In [40]: tup2[2]
```

```
Out[40]: 30
```

tuple slicing

```
In [41]: mytuple = ('one','two','three','four','five','six','seven','eight')
```

```
In [42]: mytuple[:4]
```

```
Out[42]: ('one', 'two', 'three', 'four')
```

```
In [43]: mytuple[::-1]
```

```
Out[43]: ('eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one')
```

```
In [44]: mytuple[:7]
```

```
Out[44]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven')
```

```
In [46]: mytuple[0:9:1]
```

```
Out[46]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [47]: mytuple[0:9:2]
```

```
Out[47]: ('one', 'three', 'five', 'seven')
```

remove and changes

```
In [48]: mytuple
```

```
Out[48]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [51]: mytuple[0] = 'abhi' # TUPLE IS IMMUTABLE
```

```
-----  
-----  
TypeError                                Traceback (most recent call last)  
Cell In[51], line 1  
----> 1 mytuple[0] = 'abhi' # TUPLE IS IMMUTABLE  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [52]: del mytuple[0] # deletion attribute will not work in the tuple
```

```
-----  
-----  
TypeError                                Traceback (most recent call last)  
Cell In[52], line 1  
----> 1 del mytuple[0] # deletion attribute will not work in the tuple  
  
TypeError: 'tuple' object doesn't support item deletion
```

Loop through Tuple

```
In [53]: mytuple
```

```
Out[53]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [54]: for i in mytuple:  
         print(i)
```

```
one
two
three
four
five
six
seven
eight
```

```
In [55]: for i in enumerate(mytuple):
          print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
```

```
In [56]: 'one' in mytuple
```

```
Out[56]: True
```

```
In [57]: 2 in mytuple
```

```
Out[57]: False
```

index postioning

```
In [58]: mytuple
```

```
Out[58]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight')
```

```
In [59]: mytuple.index('one')
```

```
Out[59]: 0
```

```
In [60]: mytuple.index('seven')
```

```
Out[60]: 6
```

Sorting

```
In [ ]: mytuple2 = (
```